Andrea Rovescalli

Senior software engineer

Artificial intelligence / Real-time 3D / XR

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01/07/1983 · Swiss and Italian nationality · Lausanne

Profile

Senior software engineer with 15+ years of experience at the intersection of software engineering and creative technologies. Specialised in designing and developing interactive experiences, real-time 3D applications and XR solutions, combining user interfaces, audiovisual systems, hardware and artificial intelligence. Has also held responsibilities in technical leadership, team coordination and collaboration with external partners.

Core expertise

01 **Artificial intelligence**

- Integration of large language models (LLMs) and generative AI services into applications.
- Development of LLM-based conversational applications.
- Development of multimodal indexing and semantic search solutions.
- Use of machine learning, deep reinforcement learning and imitation learning techniques.

02 **Real-time 3D & video games**

- Design and development of real-time 3D applications and video games.
- Gameplay mechanics development.
- Performance and rendering optimisation.
- Shader development.

03 **Technical leadership & coordination**

- Technical leadership and development coordination.
- Team management and coordination with external partners.
- Coordination of interdisciplinary projects.

04 **Data visualisation**

- Design and development of interactive 2D and 3D data visualisation tools.

05 **UX/UI & interaction**

- User interface (UI) design and development.

06 **XR & immersive experiences**

- Development of virtual (VR), augmented (AR) and mixed reality (MR) experiences.
- Integration of software, sensors and hardware into immersive environments.

Professional experience

EPFL+ECAL Lab | 09/2025 - present | **Software engineer**

Research software and prototypes combining AI, data visualisation and interaction design.

UDIMU SA | 02/2021 - 04/2025 | **Software engineer**

Development of real-time 3D applications and interactive systems.

Thunder Sparrow | 03/2020 - 02/2021 | **Co-founder & senior software engineer**

Co-founded the studio and developed the video game Spinny's Journey for Nintendo Switch and Steam.

OZWE SA / Oculus Studios | 07/2017 - 03/2020 | **Software engineer**

Development of virtual reality video games for mobile and PC platforms.

Ars Electronica Solutions | 01/2015 - 01/2017 | **Software engineer**

Data visualisation and interactive systems for cultural institutions and trade shows.

Equinoxe MIS Development | 11/2010 - 10/2012 | **Software engineer**

Development and maintenance of Is-Academia, an academic management system for schools and universities.

Nestlé | 07/2010 - 10/2010 · Internship | **Developer**

Web-based search and data visualisation tools for Nestlé's intranet.

EPFL | 09/2009 - 04/2010 | **Developer**

Development of content management features and web interfaces connected to backend services.

Technical skills

Artificial intelligence & machine learning	OpenAI API, PyTorch, Hugging Face Transformers, Qwen, Unity ML-Agents.
Programming languages	C++, C#, Java, JavaScript, Python, PL/SQL.
Frameworks & libraries	React, Three.js, Node.js, Vite, Zustand, Processing, jQuery, Bootstrap.
3D engines	Unity, Unreal Engine.
Shader languages	GLSL, ShaderLab.
Hardware & prototyping	Arduino, Raspberry Pi.
Audiovisual & interactive creation	Max/MSP, MadMapper.
3D modelling & creation	Blender, Substance Painter.
Development & profiling	Git, RenderDoc.

Languages

Italian: native · French: C1 · English: C1 · Spanish: B1 · German: A2

Education

2012 - 2014 | **Master's in Media and Interaction Design** | Geneva University of Art and Design (HEAD)

2008 - 2009 | **Erasmus programme** | École polytechnique fédérale de Lausanne (EPFL)

2007 - 2008 | **Master's in Computer Engineering** | University of Siena

2002 - 2007 | **Bachelor's in Computer Engineering** | Politecnico di Milano